

[Aim: 100/100 in Maths]

POLYNOMIALS $\rightarrow$ [variable]^{whole no.}

K^3b :-

Constant

$\rightarrow$ fixed value

eg 2, π - - -

mathe :- a, b, c - - -

Variables $\rightarrow$ not-fixed value

$$x - 2 = 0 \rightarrow \boxed{x = 2}$$

$$x + 1 = 0 \rightarrow \boxed{x = -1}$$

Maths $\rightarrow$ x, y, z, t - -

Algebraic Expression:- combination of constants & variables
using $+$, $-$, $\times$, $\div$.

eg $x - 4 \rightarrow \text{A.E.}$

var $\swarrow$ $\nwarrow$ const

$$x^2 - 4y + 2 \rightarrow \text{A.E.}$$

$$y^2 - \frac{5}{x} + 2$$

$$x^2 - y^2$$

$$y^2 + 2y - 6$$
$$x^{14} - 6x^{12} + 12y + t$$

$$\frac{6}{x^2} - 2y$$

$$\frac{1}{x}$$

$$xy$$

⊛ वो A.E. जिसमें [variable] whole no. , 3rd A.E. को polynomial कहा जाता है।
 $\omega = \{0, 1, 2, 3, 4, 5, \dots\}$

eg $\rightarrow x^2 - 1$ polynomial?? yes

$\rightarrow x^2 - y^4 + 2z$ yes

$\rightarrow x^2 + 4y + \frac{1}{z} \rightarrow x^2 + 4y + z^{-1}$ Not poly

$\rightarrow (x-2)^2 \Rightarrow x^2 + 4 - 4x$ yes

$\rightarrow \frac{x+1}{x^2-1} \Rightarrow \frac{x+1}{(x+1)(x-1)} \Rightarrow \frac{1}{x-1}$ Not

$\rightarrow x^3 + 4y^2 - 1$ yes

$\rightarrow t^{-1}$ Not

$\rightarrow \frac{(x-3)^3}{(x-3)} \Rightarrow \frac{(x-3)(x^2+3^2+3x)}{(x-3)} \Rightarrow x^2 + 9 + 3x$ yes

[All polynomials are A.E.
but all A.E. are not polynomials]

LP:-

$$x^3 - 4x^2 + 3x - 2$$

→ is it a polynomial? yes

→ coefficient of x^3 ? 1

→ coefficient of x^2 ? -4

→ " " x ? 3

→ constant? -2

→ degree?? 3

→ no. of terms?? 4 terms

variable & highest power available

TYPES OF POLYNOMIAL

On basis of (no. of terms):-

→ $n=1 \Rightarrow$ monomial (eg $x, y, 1, \pi, 2, \dots$)

→ $n=2 \Rightarrow$ binomial (eg $x-1, x^2+2, x^2-x, \dots$)

→ $n=3 \Rightarrow$ trinomial (eg: $x^2-x+1, x^3-x^2+x, \dots$)

On basis of degree:-

straight line

→ $D=1 \Rightarrow$ Linear $\left\{ \text{eg } x, x-2+\sqrt{2} \dots \right\}$

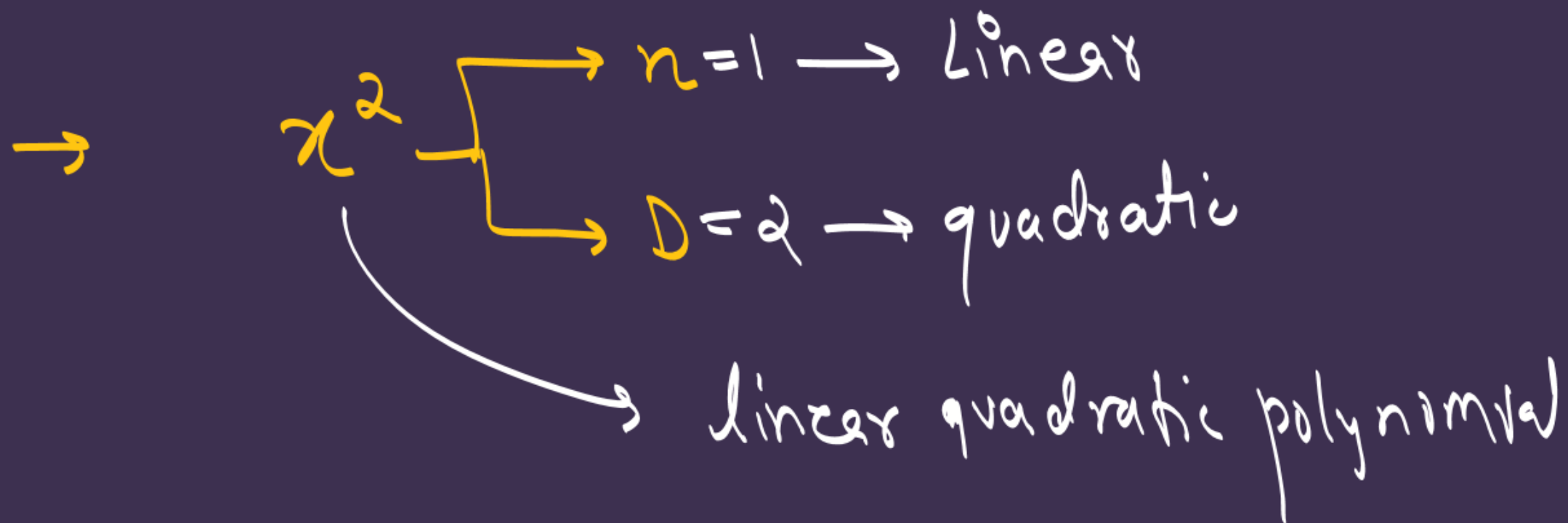
parabola



→ $D=2 \Rightarrow$ Quadratic $\left\{ \text{eg } x^2, x^2-1 \dots \right\}$

→ $D=3 \Rightarrow$ cubic $\left\{ \text{eg } x^3-x^2+1, x^3, x^3-x^2+x+3 \dots \right\}$

LP:



$\rightarrow x^2 + x - 1$
 $\left\{ \begin{array}{l} n = 3 \rightarrow \text{trinomial} \\ D = 2 \rightarrow \text{quadratic} \end{array} \right.$

$\rightarrow x^3 - x$
 $\left\{ \begin{array}{l} n = 2 \rightarrow \text{binomial} \\ D = 3 \rightarrow \text{cubic} \end{array} \right.$

$\rightarrow x^3 + x^{-2}$
 $x^3 + \frac{1}{x^2}$
 $\left\{ \begin{array}{l} n = 2 \rightarrow \text{binomial} \\ D = 3 \rightarrow \text{cubic} \end{array} \right.$
 $\rightarrow \text{cubic binomial}$

Polynomial X

Constant Polynomial: $2, \pi, -1, 3, \dots$

→ degree of constant poly $\Rightarrow 2x^0 \rightarrow 2 \times 1$
 $\Rightarrow \boxed{D=0}$ ✓

Zero Polynomial: $0x^0 / 0x^1 / 0x^2 / 0x^3 \dots$
 $\Rightarrow 0 \quad \Rightarrow 0 \quad \Rightarrow 0 \quad \Rightarrow 0$

→ degree of zero poly $\Rightarrow \underline{\text{not defined}}$ ✓

(Hw) $\rightarrow$ $\frac{\text{Next DPP.}}{\text{Module.}}$ | (II) $\rightarrow$ Q1

Complex Real No. module